Prerequisites Quiz

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Not for a grade. Answer what you can, and write "not sure" instead of guessing. That tells me more than a lucky answer does.

1. Rate yourself on each line, where 1 is not at all and 5 is extremely familiar. Circle one per line.

a. Using GNU/Linux from a shell prompt ..... 1 ● 3 4 5

b. Writing C, as opposed to C++ ..... 1 ● 3 4 5

c. Reading a man page to find what a function returns on failure ..... 1 ● 3 4 5

2. Assume a 64-bit Linux machine. Fill in the value of each expression. Where the value is an address, say which element it points to.

```c
int  a[5] = { 10, 20, 30, 40, 50 };
int *p    = a;
```

Expression	Value
`p[3]`	40
`*(p + 2)`	points to next iterative value
`*p + 2`	whichever value p is currently pointing at plus 2
`p + 2`	address value increased by one iteration
`&a[4] - p`	address of a[4] minus the address of p
`sizeof(a)`	5
`sizeof(p)`	would give size of address

3. Strings.

```
char  s1[] = "goblin";
char *s2   = "goblin";
char  s3[6];

strcpy(s3, s1);
```

a. `sizeof(s1)` = 6 `strlen(s1)` = 6 `sizeof(s2)` = 6

b. One line above will corrupt memory. Which one, and why?

strcpy(s3, s1)

c. The line `s2[0] = 'G';` compiles cleanly. What happens when it runs, and why?

got no clue

4. This function compiles (with a warning) and then misbehaves at run time.

```
char *make_greeting(const char *name)
{
    char buf[64];

    snprintf(buf, sizeof(buf), "Hello, %s!", name);
    return buf;
}
```

a. What is wrong with it?

i hope you find my many ways of saying idk entertaining

b. Give two different ways to fix it. For each, say who is responsible for releasing the memory.

no clue

5. For each item below, say where it lives in the process and when it stops being valid.

```c
int total;                  /* file scope */

void tally(int n)
{
    int         local = n;
    static int  calls = 0;
    const char *label = "count";
    int        *data  = malloc(n * sizeof *data);

    calls++;
}
```

Item	Where it lives	When it becomes invalid
`total`	yeah i got no idea	
`local`		
`calls`		
`the characters "count"`		
`the block data points to`		

6. Assume n is 10 and the declarations are in scope. There are four separate defects in this fragment. List them.

```c
int *v = malloc(n);

for (int i = 0; i <= n; i++)
    v[i] = i * i;

free(v);
printf("first = %d\n", v[0]);
```

i would answer. But i would just be making an estimated guess

7. Structs.

```
struct point { int x, y; };

void shift(struct point p, int dx, int dy)
{
    p.x += dx;
    p.y += dy;
}
```

a. A caller does `shift(pt, 1, 2);` and finds that `pt` is unchanged. Why?

The function does not point to the orginal value entered. It just creates values inside itself.

So no outside values are actually affected by it.

b. Rewrite `shift()` so that it does modify the caller's point. Write the full signature and body in the box.

```
struct point { int x, y; };

void shift(struct point *p, int dx, dy){
p.x += dx;
p.y += dy;
}
```

c. If pp is a `struct point *`, write two equivalent expressions that read the x field, one per box.

idk

idk

8. `path` names a readable file.

```
int  fd = open(path, O_RDONLY);
char buf[100];

read(fd, buf, 100);
printf("%s\n", buf);
```

a. Name three things this code fails to check or handle.

idk

b. Why is `printf("%s", buf)` the wrong way to print what `read()` produced?

idk

9. Reporting failure.

a. A system call such as `open()` fails. What value does it return, and where does the reason for the failure go?

idk

b. A student calls `printf()`, then tests `errno`, sees that it is non-zero, and reports an error. Explain why that reasoning is wrong.

idk

c. What value does each of these return to signal failure?

Function	Returns on failure
`malloc()`	idk
`fopen()`	idk
`fork()`	idk
`getchar()`	idk

10. Exit status.

a. What does it mean for `main()` to return 0? What does a non-zero return mean?

Everything in main finished running.

Non zero return implies some sort of memory leak or unfinihsed task

b. You have just run `./a.out` at a bash prompt. What do you type to see the exit status it returned?

don't know